

SAMRIDHA SHRESTHA

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SUMMARY

Computer Engineering graduate and software developer with hands-on experience building production web platforms and AI-driven tools, alongside a strong analytical background in deep learning-based data analysis. Comfortable leveraging AI tools and automation to work efficiently, with close attention to detail, structured documentation, and clear written and verbal communication.

EDUCATION

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT (T.U AFFILIATED)

Kalanki, Kathmandu

Bachelor in Computer Engineering 2021-2026

ST. XAVIER'S COLLEGE

Maitighar, Kathmandu

Plus Two 2019-2021

EXPERIENCE

FULL STACK DEVELOPER INTERN June 2026 - Present

Octacore Solutions

- Contributing to full-stack development of creationtextile.com.np, a Django and Next.js e-commerce platform, building and maintaining features across backend APIs and frontend UI.
- Built a chatbot using Retrieval-Augmented Generation (RAG) to support customer interactions on the platform, using hybrid search (BM25 + vector embeddings), a reranker, prompt engineering, and conversation history handling.
- Building a chatbot for cheers.com.np using n8n for workflow automation and orchestration.

FREELANCE WEB DEVELOPER 2026

Aqua Hundred (Aqua Minerals Nepal Pvt. Ltd.) (aquahundred.com.np)

- Designed and developed a full multi-page marketing website for Aqua Hundred (Aqua Minerals Nepal Pvt. Ltd.), featuring scroll-driven animations, an infinite product carousel, multi-step order modal, a Leaflet.js depot map, and Facebook Messenger integration; deployed to Cloudflare Pages with sitemap, robots.txt, and CSP configuration.

PROJECTS

AUTOMATED LANDFILL SITE

SUITABILITY MAPPING USING DEEP LEARNING 2025-2026

- A landfill site suitability map of Kathmandu valley and its neighboring districts was generated on geospatial data using segmentation deep learning models like Unet, ResUnet and DeeplabV3.
- Custom U-Net achieved best performance with 96.97% accuracy, 0.92 mean IoU and 0.96 Dice score.

FLOOD SUSCEPTIBILITY MAPPING OF

KATHMANDU VALLEY USING DEEP LEARNING 2024-2025

- A deep learning approach for risk assessment and flood susceptibility mapping due to urban fluvial floods in the Kathmandu Valley using CNN and RF on geospatial data.
- CNN achieved 95% accuracy with balanced class performance, outperforming Random Forest.

SKILLS

Technical Skills: Python, Html/CSS, JavaScript, Django, Next.js/React, PyTorch, Scikit-Learn, RAG/LLM Systems (hybrid search, vector databases, rerankers), n8n, Git, C, C++, QGIS, SQL, Figma
Soft Skills: Problem Solving, Analytical Mindset, Attention to Detail, Written & Verbal Communication, Teamwork, Time Management